John Francis Aradan

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Education

University of Wisconsin-Madison **Madison, WI**
BS in Computer Sciences, CERT Leadership ***GPA: 3.72*** *Expected Dec 2026*
Transferred from ICAS, Manipal (GPA: 3.87, Rank 2 of 2024 cohort)

Awards

- **Sophomore Research Fellowship, UW–Madison –** Funded ML-based cancer screening research at Striker Lab.
- **OpenAI Research Stipend, UW–Madison / OpenAI –** Funded developing RAG based LLM Model for VetAI.

Experiences

PriceRight Labs (Personal Product) **Madison, WI**
End to End ML Developer *06/2025 - Present*
- Built a full-stack dynamic pricing system for local events, scraping & processing 2,500+ listings across 6 major cities with >98% parsing accuracy and structured + NLP-derived feature extraction0.
- Trained and evaluated pricing models achieving RMSE < 26, then developed a dynamic adjustment layer (demand, time-to-event, weather) that increased simulated revenue by 12–25%.
- Designed an interactive Streamlit dashboard with real-time pricing, Price explainability, and scenario simulation.

University of Wisconsin - Madison **Madison, WI**
Undergraduate Research Assistant – ***Striker Lab*** *08-2024 - Present*
- Developed ML models on 2,000+ patient records to predict HPV progression stages; RF achieved 83% accuracy.
- Performed robust preprocessing: conditional filtering, one-hot encoding, & outlier handling on noisy clinical datasets.
- Visualized model behavior via confusion matrices and custom accuracy plots

VetAI **Madison, WI**
End to End ML Developer *06/2025-08/2025*
- Designed a RAG pipeline indexing 1200+ docs via OpenAI embeddings & Pinecone, gaining ≤400ms query latency.
- Built a full-stack Streamlit chatbot with GPT-4 summarization and SendGrid integration; increased session durations 2× and achieved 82% open rate on email summaries.
- Integrated OpenAI Moderation API pre/post-processing, resulting in zero policy violations across 300+ interactions
- Added auto-learn module to handle knowledge gaps and enrich Pinecone vectors, enabling continuous self-learn.

University of Nebraska – Omaha (in collaboration with **Stanford Med.)** **Omaha, NE**
Researcher *04/2023 – 11/2023*
- Trained UNet-based CNNs with pretrained encoders (VGG16, ResNet50, InceptionNet) for eye lesion segmentation.
- Achieved 0.81 Dice coefficient on validation set using grayscale-enhanced images and aggressive augmentation.
- Labeled and preprocessed 500+ scans with binary masks and optimized inference for smooth region boundaries.

Projects

2023 PennApps Project – University of Pennsylvania (UPenn)
- Built diagnostic web app using Flask and InceptionResNetV2 to classify dermaal diseases from user-uploaded images.
- Integrated structured medical history into custom rule-based classifier to enhance severity prediction.
- Deployed working prototype on Porkbun; won awards in image processing and NLP tracks.

Lung Tumor Segmentation via Deep UNet Architectures
- Trained binary UNet models with VGG16, DenseNet121, and InceptionResNetV2 encoders to segment lung nodules; applied grayscale normalization, histogram equalization, and elastic deformation to augment limited CT dataset.
- Achieved Dice = 0.79 and IoU = 0.74; built mask evaluation pipeline to visualize overlays and boundary metrics.

Research Publications

- "Prediction of Cardiovascular Diseases using ML Algorithms" – INOCON 2023, IEEE
- "ML Techniques for Multiclass Brain Tumor Classification" – RCAAI 2023, Springer Nature

Technical Skills

Languages: Python, MATLAB, C++, SQL
ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, OpenCV, CNNs, Transfer Learning, Model Evaluation
LLMs & Retrieval: Pinecone, Prompt Engineering, RAG, LangChain,
Tools & Platforms: Streamlit, AWS, Git, PostgreSQL, SendGrid, pandas, NumPy, Matplotlib, W&B, Docker